Stat 380 Mini Project 2

Ashley Song, Ian Peralta, Bryan Xiao

Libraries and data load

Libraries for data work and plotting, and for the models that follow.

```
library(tidyverse)
library(ggplot2)
library(parsnip)
library(rsample)
library(recipes)
library(workflows)
library(tune)
library(dials)
library(yardstick)
library(broom)
```

The response is `Democracy`, whether a country was a democracy in 2020. Every predictor is taken for 2020, the year of the democracy data.

```
sanitation <- read_csv("../data/predictors/at_least_basic_sanitation_overall_access_percent.
fertility  <- read_csv("../data/predictors/children_per_woman_total_fertility.csv")
co2        <- read_csv("../data/predictors/co2_pcap_cons.csv")
income     <- read_csv("../data/predictors/ilevels4_wb.csv", col_types = cols(.default = "c")
poverty    <- read_csv("../data/predictors/number_of_people_in_poverty.csv")
democracy <- read_csv("../data/response/democracy2020.csv")
```

Clean data: drop the years after 2022 and change each file to long format, one row per country and year.

```r
drop_and_pivot <- function(df, value_name) {
  df |>
    select(-any_of(as.character(2023:2100))) |>
    pivot_longer(
      cols = -c(geo, name),
      names_to = "year",
      values_to = value_name
    ) |>
    mutate(year = as.integer(year))
}

sanitation_long <- drop_and_pivot(sanitation, "sanitation_access_pct")
fertility_long  <- drop_and_pivot(fertility, "children_per_woman")
co2_long        <- drop_and_pivot(co2, "co2_per_capita")
income_long     <- drop_and_pivot(income, "income_level")
poverty_long    <- drop_and_pivot(poverty, "people_in_poverty")
```

Join variables into one data frame: keep 2020, add the response, and drop countries with any missing value.

```r
joined <- list(
  sanitation_long, fertility_long, co2_long,
  income_long, poverty_long
) |>
  reduce(full_join, by = c("geo", "name", "year")) |>
  filter(year == 2020) |>
  arrange(geo, year) |>
  inner_join(democracy, by = c("geo", "name"))

combined <- na.omit(joined)
write.csv(combined, "../data/combined_data.csv")
```

Countries in the data

```r
joined |>
  summarise(across(c(co2_per_capita, income_level, sanitation_access_pct,
                     people_in_poverty), \(x) sum(is.na(x)))) |>
  knitr::kable(caption = "Missing values among countries with a democracy label")
```

Table 1: Missing values among countries with a democracy label

co2_per_capita	income_level	sanitation_access_pct	people_in_poverty
0	0	5	6

```
joined |>
  count(Democracy, name = "all_countries") |>
  left_join(combined |> count(Democracy, name = "kept"), by = "Democracy") |>
  knitr::kable(caption = "Countries before and after dropping rows with missing values")
```

Table 2: Countries before and after dropping rows with missing values

Democracy	all_countries	kept
No	91	84
Yes	74	70

```
anti_join(joined, combined, by = "geo") |>
  select(name, Democracy, income_level, co2_per_capita) |>
  knitr::kable(digits = 2, caption = "Countries dropped for missing values")
```

Table 3: Countries dropped for missing values

name	Democracy	income_level	co2_per_capita
UAE	No	High income	19.74
Argentina	Yes	Upper middle income	3.52
Azerbaijan	No	Upper middle income	3.79
Bosnia and Herzegovina	No	Upper middle income	6.15
Eritrea	No	Low income	0.22
Equatorial Guinea	No	Upper middle income	4.35
Luxembourg	Yes	High income	12.57
Nicaragua	No	Lower middle income	0.86
Qatar	No	High income	25.63
Slovenia	Yes	High income	8.85
Trinidad and Tobago	Yes	High income	17.16

The democracy file has 165 countries and 154 remain after dropping rows with a missing value. CO2 and income level are never the missing ones, since every dropped country lacks a sanitation or poverty value. Two high-CO2 non-democracies, Qatar and the UAE, are among the dropped countries, so all results below describe the 154 kept.

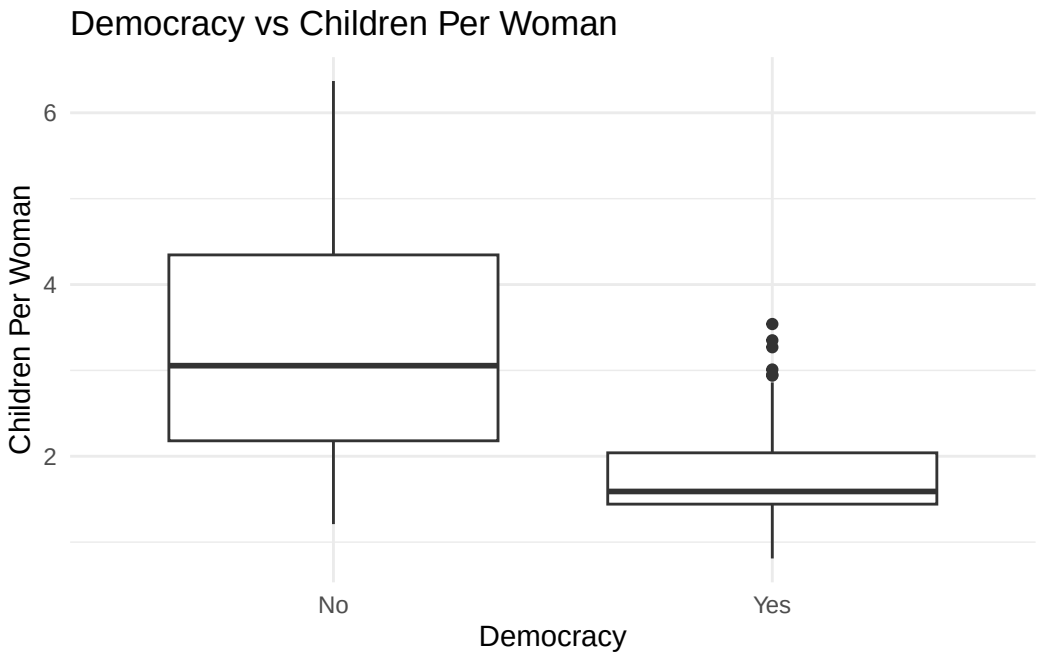
Exploratory data analysis

Each predictor is graphed against the response, `Democracy`, and summarized by democracy group. Quantitative predictors get summary statistics and an outlier check, and the categorical income group gets a two way table.

Variable 1: Children Per Woman

Graph and description

```
combined |>
  ggplot(mapping = aes(x = Democracy, y = children_per_woman)) +
  geom_boxplot() +
  labs(title = "Democracy vs Children Per Woman",
       x = "Democracy", y = "Children Per Woman") +
  theme_minimal()
```



Non-democratic countries generally have more children per woman, with a median of about 3.06 compared to 1.59 in democratic countries. Non-democratic countries also show greater variability.

Stats and description

```
combined |>
  summarise(
    n_countries = n(),
    mean        = mean(children_per_woman),
    median      = median(children_per_woman),
    sd          = sd(children_per_woman),
    min         = min(children_per_woman),
    q1          = quantile(children_per_woman, 0.25),
    q3          = quantile(children_per_woman, 0.75),
    max         = max(children_per_woman),
    .by = Democracy
  )
```

```
# A tibble: 2 x 9
  Democracy n_countries  mean median    sd   min    q1    q3   max
  <chr>           <int> <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 No                 84  3.32   3.06 1.34   1.21  2.18  4.34  6.37
2 Yes                70  1.80   1.59 0.578  0.81  1.44  2.04  3.54
```

1. Non-democratic countries have a higher mean and median children per woman than democratic countries. They also have a larger standard deviation, indicating more variation. Six democracies are extreme outliers, led by Ghana at 3.54 children per woman, Namibia at 3.35 and Papua New Guinea at 3.27.

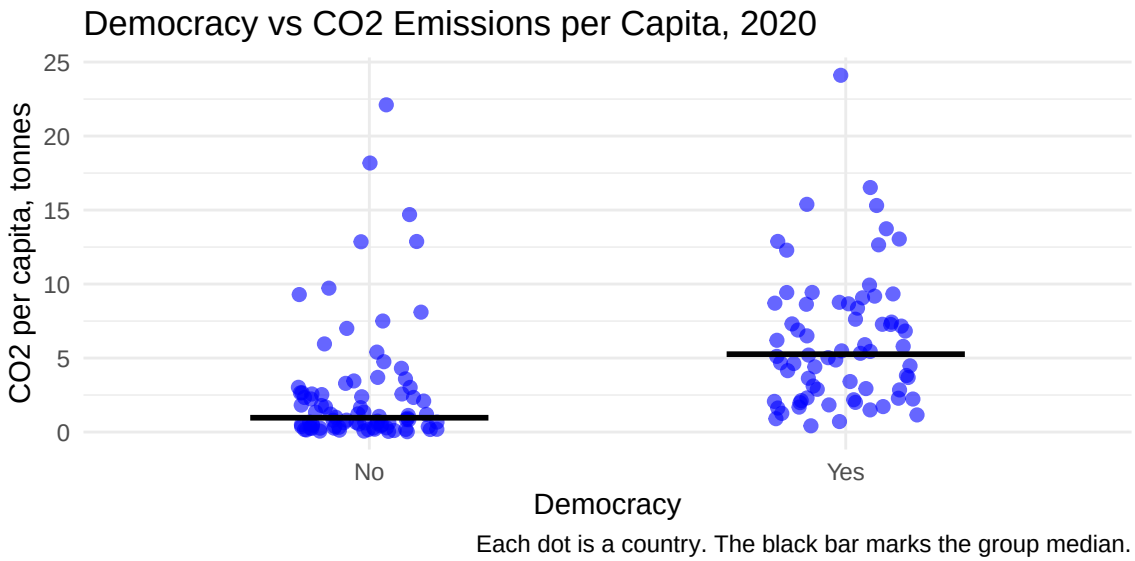
Variable 2: CO2 Emissions Per Capita

`co2_per_capita` is quantitative and continuous, measured in tonnes of CO2 per person on a consumption basis. In 2020 the kept countries range from 0.02 to 24.1 tonnes, with no negative or missing values.

Graph and description

```
ggplot(data = combined, mapping = aes(x = Democracy, y = co2_per_capita)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.6, color = "blue", size = 2) +
  stat_summary(fun = median, geom = "crossbar", width = 0.5,
               color = "black", linewidth = 0.4) +
  labs(title = "Democracy vs CO2 Emissions per Capita, 2020",
```

```
       x = "Democracy", y = "CO2 per capita, tonnes",
       caption = "Each dot is a country. The black bar marks the group median.") +
  theme_minimal()
```



Democratic countries emit more CO2 per person. Most non-democratic countries sit below 3 tonnes and half are below about 1 tonne, while democratic countries spread from about 0.4 to 24 tonnes. A few non-democracies reach far above the rest of their group.

Stats and description

```
combined |>
  group_by(Democracy) |>
  summarize(
    n_countries = n(),
    mean        = mean(co2_per_capita),
    median      = median(co2_per_capita),
    sd          = sd(co2_per_capita),
    min         = min(co2_per_capita),
    q1          = quantile(co2_per_capita, 0.25),
    q3          = quantile(co2_per_capita, 0.75),
    max         = max(co2_per_capita)
  ) |>
  knitr::kable(digits = 2, caption = "CO2 per capita by democracy")
```

Table 4: CO2 per capita by democracy

Democracy	n_countries	mean	median	sd	min	q1	q3	max
No	84	2.63	0.97	4.11	0.02	0.37	2.65	22.11
Yes	70	6.18	5.26	4.49	0.42	2.44	8.66	24.11

```
combined |>
  group_by(Democracy) |>
  filter(co2_per_capita > quantile(co2_per_capita, 0.75) + 1.5 * IQR(co2_per_capita)) |>
  arrange(Democracy, desc(co2_per_capita)) |>
  ungroup() |>
  select(Democracy, name, co2_per_capita) |>
  knitr::kable(digits = 2, caption = "Extreme CO2 outliers within each group")
```

Table 5: Extreme CO2 outliers within each group

Democracy	name	co2_per_capita
No	Kuwait	22.11
No	Saudi Arabia	18.18
No	Bahrain	14.69
No	Turkmenistan	12.88
No	Oman	12.86
No	Kazakhstan	9.72
No	Russia	9.29
No	Libya	8.10
No	Iran	7.50
No	China	7.00
Yes	Singapore	24.11

```
co2_medians <- combined |>
  group_by(Democracy) |>
  summarise(m = median(co2_per_capita)) |>
  deframe()

tibble(
  statement = c("Democracies above the non-democratic median",
                "Non-democracies above the democratic median"),
  share = c(mean(combined$co2_per_capita[combined$Democracy == "Yes"] > co2_medians["No"]),
            mean(combined$co2_per_capita[combined$Democracy == "No"] > co2_medians["Yes"]))
```

```
) |>
  knitr::kable(digits = 3, caption = "Overlap between the two groups")
```

Table 6: Overlap between the two groups

statement	share
Democracies above the non-democratic median	0.957
Non-democracies above the democratic median	0.143

Democratic countries have a mean of 6.18 and a median of 5.26 tonnes, against a mean of 2.63 and a median of 0.97 for non-democratic countries. The non-democratic mean is pulled up by a few heavy emitters, and the spread is similar in both groups, with standard deviations of 4.11 and 4.49. Among democracies 96% emit more than the non-democratic median, while only 14% of non-democracies emit more than the democratic median.

Ten non-democracies and one democracy are extreme outliers. The non-democratic ones are led by Kuwait at 22.1 tonnes, Saudi Arabia at 18.2 and Bahrain at 14.7, followed by Turkmenistan, Oman, Kazakhstan, Russia, Libya, Iran and China. Most are oil and gas exporters, and China and Russia are very large industrial economies. The one democratic outlier is Singapore at 24.1, a small wealthy trade hub. The lowest emitters are the DR Congo, Burundi and the Central African Republic, all non-democracies, and the lowest democracy is Timor-Leste at 0.42.

CO2 per capita is a useful predictor, since the middle halves of the two groups barely overlap, running from 0.37 to 2.65 tonnes for non-democracies and from 2.44 to 8.66 for democracies. Its main weakness is the group of oil and gas states, which are non-democratic but emit as much as rich democracies.

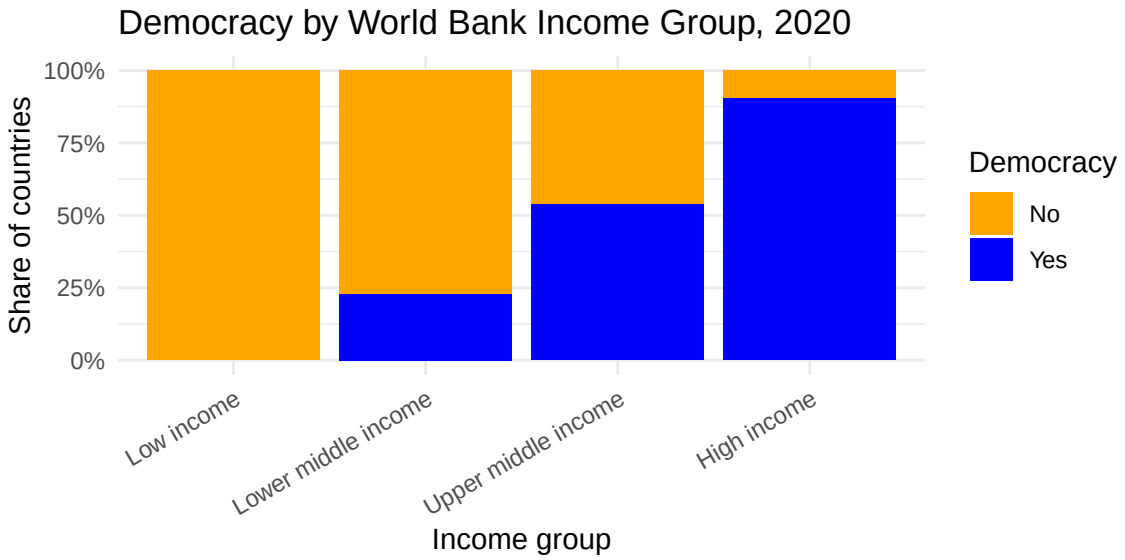
Variable 3: Income Groups

`income_level` is categorical and ordinal with four levels: Low, Lower middle, Upper middle and High income, assigned by the World Bank from GNI per capita. It has no missing values among the kept countries, and it is ordered below so that plots and tables run from low to high.

```
income_data <- combined |>
  mutate(income_level = factor(
    income_level,
    levels = c("Low income", "Lower middle income",
               "Upper middle income", "High income"),
    ordered = TRUE
  ))
```

Graph and description

```
ggplot(data = income_data, mapping = aes(x = income_level, fill = Democracy)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c(No = "orange", Yes = "blue")) +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Democracy by World Bank Income Group, 2020",
       x = "Income group", y = "Share of countries", fill = "Democracy") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



The share of democracies rises steadily with income. No low income country is a democracy, about a quarter of lower middle income countries are, about half of upper middle income countries are, and nearly all high income countries are.

Two way table and description

```
income_table <- table(income_data$income_level, income_data$Democracy)

knitr::kable(addmargins(income_table),
             caption = "Number of countries by income group and democracy")
```

Table 7: Number of countries by income group and democracy

	No	Yes	Sum
Low income	24	0	24
Lower middle income	37	11	48
Upper middle income	19	22	41
High income	4	37	41
Sum	84	70	154

```
knitr::kable(round(prop.table(income_table, margin = 1), 3),
             caption = "Share of each income group that is democratic or not")
```

Table 8: Share of each income group that is democratic or not

	No	Yes
Low income	1.000	0.000
Lower middle income	0.771	0.229
Upper middle income	0.463	0.537
High income	0.098	0.902

```
knitr::kable(round(prop.table(income_table, margin = 2), 3),
             caption = "Share of each democracy group in each income level")
```

Table 9: Share of each democracy group in each income level

	No	Yes
Low income	0.286	0.000
Lower middle income	0.440	0.157
Upper middle income	0.226	0.314
High income	0.048	0.529

Of the 154 countries, 45% are democracies. The share is 0% of 24 low income countries, 23% of 48 lower middle income countries, 54% of 41 upper middle income countries and 90% of 41 high income countries. Looking the other way, 53% of the 70 democracies are high income and none are low income, while 44% of the 84 non-democracies are lower middle income, 29% are low income and only 5% are high income.

The four high income non-democracies are Bahrain, Kuwait, Oman and Saudi Arabia, all Gulf oil exporters. Qatar and the UAE, dropped above for a missing poverty value, fit the same pattern.

```r
income_data |>
  group_by(income_level) |>
  summarize(n_countries = n(), median_co2 = median(co2_per_capita),
            mean_co2 = mean(co2_per_capita)) |>
  knitr::kable(digits = 2, caption = "CO2 per capita by income group")
```

Table 10: CO2 per capita by income group

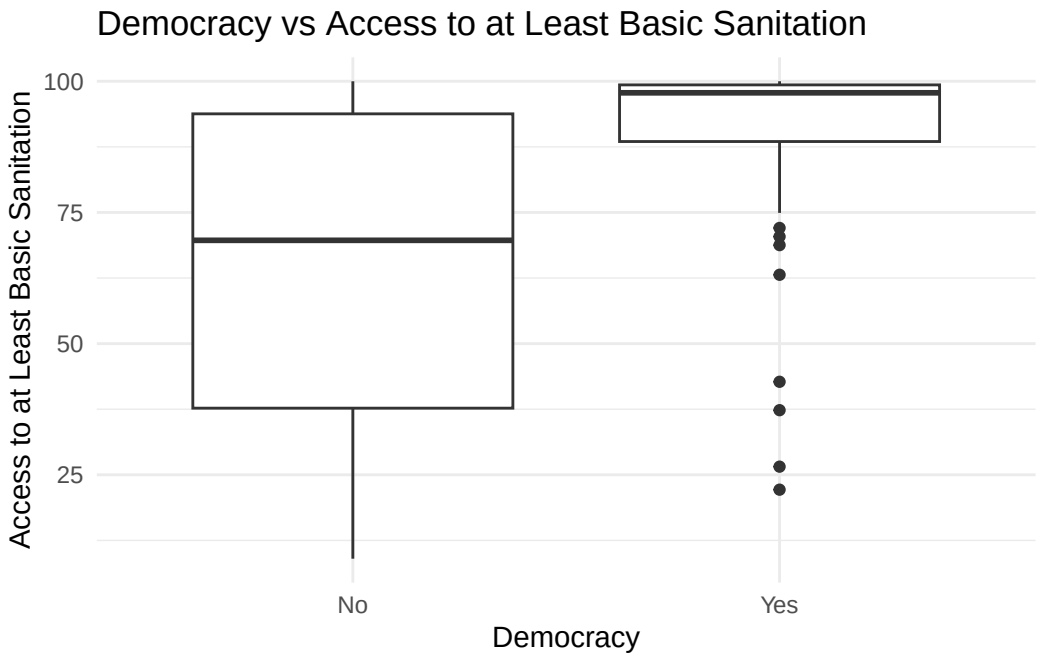
income_level	n_countries	median_co2	mean_co2
Low income	24	0.19	0.34
Lower middle income	48	1.03	1.67
Upper middle income	41	3.41	4.07
High income	41	8.67	9.72

CO2 per capita rises with income, from a median of 0.19 tonnes for low income countries to 8.67 for high income countries. The two predictors therefore capture overlapping information.

Variable 4: Basic Sanitation Access Overall

Graph and description

```r
combined |>
  ggplot(mapping = aes(x = Democracy, y = sanitation_access_pct)) +
  geom_boxplot() +
  labs(title = "Democracy vs Access to at Least Basic Sanitation",
       x = "Democracy", y = "Access to at Least Basic Sanitation") +
  theme_minimal()
```



Countries with democracy generally have higher access to basic sanitation. The median is around 98% for democratic countries compared to 70% for non-democratic countries.

Stats and description

```
combined |>
  summarise(
    n_countries = n(),
    mean        = mean(sanitation_access_pct),
    median      = median(sanitation_access_pct),
    sd          = sd(sanitation_access_pct),
    min         = min(sanitation_access_pct),
    q1          = quantile(sanitation_access_pct, 0.25),
    q3          = quantile(sanitation_access_pct, 0.75),
    max         = max(sanitation_access_pct),
    .by = Democracy
  )
```

```
# A tibble: 2 x 9
  Democracy n_countries  mean median    sd   min    q1    q3   max
  <chr>           <int> <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 No                 84  65.0   69.7  29.9  9.01  37.7  93.8   100
```

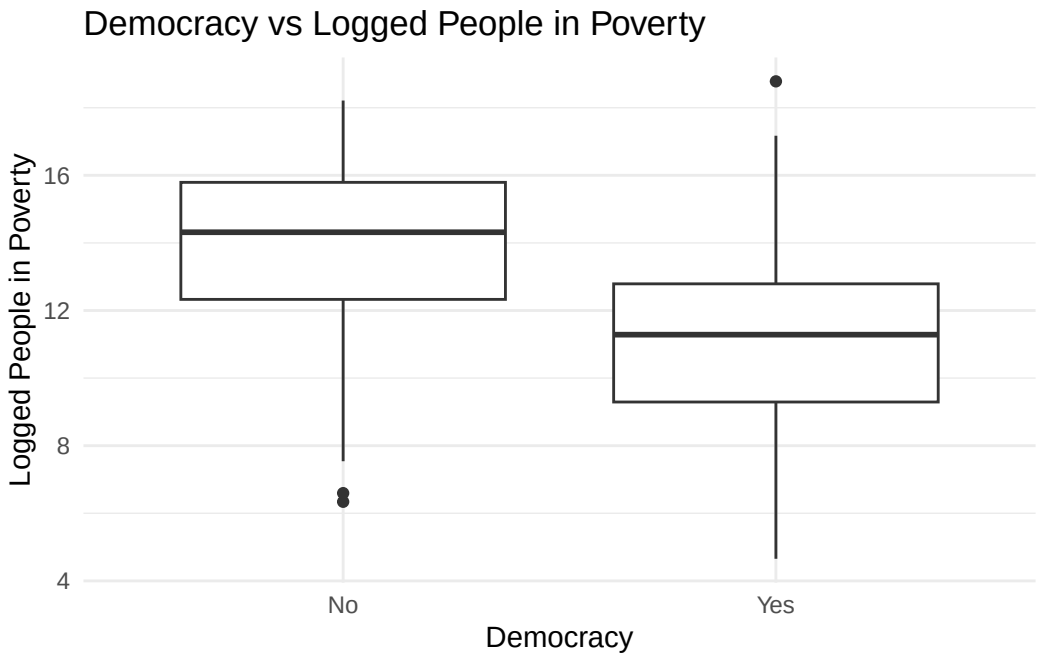
```
2 Yes                70  89.9   97.8  16.8 22.2   88.5  99.3   100
```

Democratic countries have a higher mean (89.89%) and median (97.80%) of access to basic sanitation than non-democratic countries (mean 64.98%, median 69.69%). Eight democracies are extreme low outliers, the lowest being Papua New Guinea at 22.2%, Ghana at 26.6% and Namibia at 37.3%.

Variable 5: Number of People in Poverty

Graph and description

```r
combined <- combined |>
  mutate(log_people_in_poverty = log(people_in_poverty))
combined |>
  ggplot(mapping = aes(x = Democracy, y = log_people_in_poverty)) +
  geom_boxplot() +
  labs(title = "Democracy vs Logged People in Poverty",
       x = "Democracy", y = "Logged People in Poverty") +
  theme_minimal()
```



Non-democratic countries generally have higher logged poverty values, with a median around 14 compared to approximately 11 for democratic countries.

Stats and description

```
combined |>
  summarise(
    n_countries = n(),
    mean        = mean(people_in_poverty),
    median      = median(people_in_poverty),
    sd          = sd(people_in_poverty),
    min         = min(people_in_poverty),
    q1          = quantile(people_in_poverty, 0.25),
    q3          = quantile(people_in_poverty, 0.75),
    max         = max(people_in_poverty),
    .by = Democracy
  )
```

```
# A tibble: 2 x 9
  Democracy n_countries     mean  median        sd   min      q1       q3    max
  <chr>           <int>    <dbl>   <dbl>     <dbl> <dbl>   <dbl>    <dbl>  <dbl>
1 No                 84 7420427. 1654098 14608717.   567 226138. 7216626  8.12e7
2 Yes                70 3710988.   79680 17626321.   105  10856.  362742. 1.43e8
```

Non-democratic countries have a higher mean and median number of people in poverty than democratic countries, although the large standard deviation shows substantial variation within both groups. On the raw scale 24 countries are high outliers, 10 non-democracies led by the DR Congo at 81.2 million and Nigeria at 80.2 million, and 14 democracies led by India at 143.3 million and Indonesia at 28.7 million.

Which variable for single variable models?

Each predictor is scored with two rough measures. Accuracy is the share of countries classified correctly by the best single cutoff on that predictor. AUC is the chance that a randomly chosen democracy scores higher than a randomly chosen non-democracy on the predictor, taken in whichever direction is stronger. A value of 0.5 means no separation and 1 means perfect separation.

```
is_democracy <- combined$Democracy == "Yes"

rank_auc <- function(x, y) {
  r <- rank(x)
  n_yes <- sum(y)
```

```r
  n_no <- sum(!y)
  auc <- (sum(r[y]) - n_yes * (n_yes + 1) / 2) / (n_yes * n_no)
  max(auc, 1 - auc)
}

best_cutoff_accuracy <- function(x, y) {
  cutoffs <- sort(unique(x))
  max(map_dbl(cutoffs, \(t) max(mean((x > t) == y), mean((x <= t) == y))))
}

income_rank <- match(
  combined$income_level,
  c("Low income", "Lower middle income", "Upper middle income", "High income")
)

predictors <- list(
  "Children per woman"       = combined$children_per_woman,
  "CO2 emissions per capita" = combined$co2_per_capita,
  "Income group"             = income_rank,
  "Basic sanitation access"  = combined$sanitation_access_pct,
  "People in poverty"        = combined$people_in_poverty
)

tibble(
  predictor = names(predictors),
  accuracy  = map_dbl(predictors, best_cutoff_accuracy, y = is_democracy),
  auc       = map_dbl(predictors, rank_auc, y = is_democracy)
) |>
  arrange(desc(auc)) |>
  knitr::kable(digits = 3, caption = "Single predictor strength for separating democracies")
```

Table 11: Single predictor strength for separating democracies

predictor	accuracy	auc
Income group	0.779	0.859
Children per woman	0.792	0.855
CO2 emissions per capita	0.753	0.809
Basic sanitation access	0.708	0.773
People in poverty	0.734	0.746

```r
combined |>
  select(children_per_woman, co2_per_capita, sanitation_access_pct, people_in_poverty) |>
  cor() |>
  knitr::kable(digits = 2, caption = "Correlations between the quantitative predictors")
```

Table 12: Correlations between the quantitative predictors

	children_per_woman	co2_per_capita	sanitation_access_pct	people_in_poverty
children_per_woman	1.00	-0.53	-0.85	0.34
co2_per_capita	-0.53	1.00	0.57	-0.25
sanitation_access_pct	-0.85	0.57	1.00	-0.36
people_in_poverty	0.34	-0.25	-0.36	1.00

Income group and children per woman are nearly tied as the strongest predictors, with AUCs of about 0.86 each, and CO2 comes next at 0.81. Sanitation and poverty are the weakest. We suggest children per woman for the single variable models, because it has the best accuracy at 79% and is quantitative with many distinct values, which suits both kNN distances and a logistic curve. Income group has only four distinct values, which makes it awkward for kNN. Children per woman and sanitation are strongly related, with a correlation of -0.85, so the all variable models will carry some duplicated information. These are rough checks on the same data used to build them, and cross validation in the models will test them properly.

kNN classification

Predictors are normalized so distances are fair, and k is tuned from 3 to 21 with 10 fold cross validation using the same seed as the logistic models. Training accuracy is a little optimistic for kNN because each country counts as its own neighbor.

Single variable model

Make a kNN model using your best predictor variable. This model uses children per woman, the best predictor from the exploratory analysis.

```
knn_single_variable_recipe <- recipe(Democracy ~ children_per_woman, data = combined) |>
  step_normalize(all_numeric_predictors())

knn_all_variable_recipe <- recipe(Democracy~income_level+sanitation_access_pct+children_per_w
  step_normalize(all_numeric_predictors())|>
  step_dummy(all_nominal_predictors())
```

Model tuning to find k

```
set.seed(1776)
folds <- vfold_cv(combined, v = 10)

knn_spec <- nearest_neighbor(neighbors = tune(), weight_func = "rectangular") |>
  set_engine("kknn") |>
  set_mode("classification")

knn_wf_single <- workflow() |>
  add_recipe(knn_single_variable_recipe) |>
  add_model(knn_spec)

knn_grid <- tibble(neighbors = seq(from = 3, to = 21, by = 2))
class_met <- metric_set(accuracy)

knn_cv_single <- tune_grid(
  knn_wf_single,
  resamples = folds,
  grid = knn_grid,
  metrics = class_met
)
```

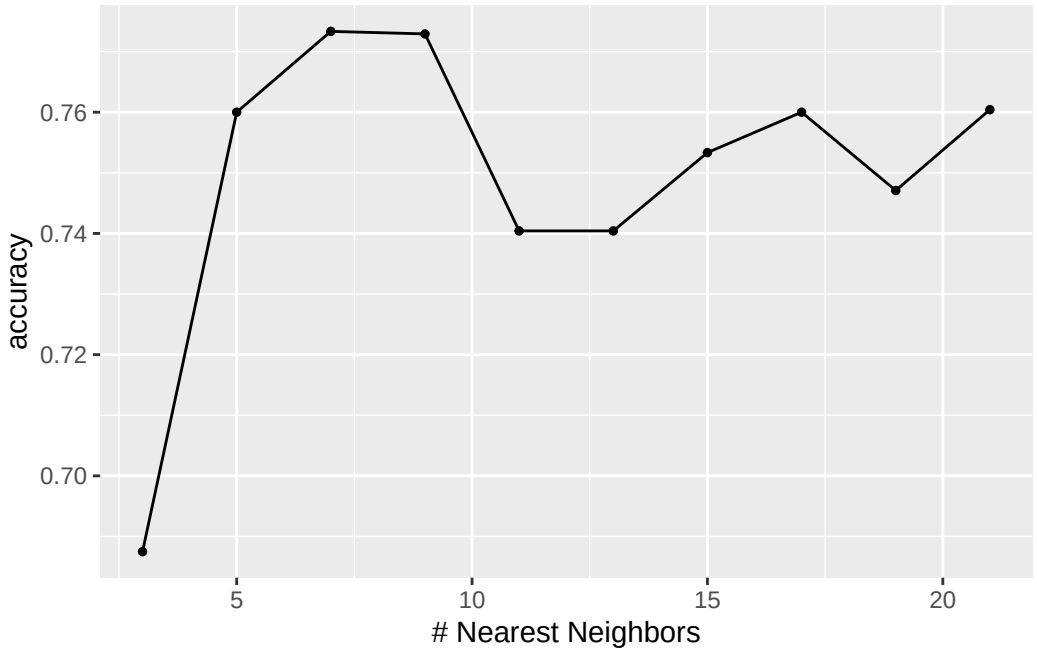
```
> A | warning: package 'kknn' was built under R version 4.5.3

There were issues with some computations   A: x1
There were issues with some computations   A: x1
```

```
collect_metrics(knn_cv_single)
```

```
# A tibble: 10 x 7
   neighbors .metric  .estimator  mean     n std_err .config
       <dbl> <chr>    <chr>      <dbl> <int>   <dbl> <chr>
 1         3 accuracy binary     0.688    10  0.0357 pre0_mod01_post0
 2         5 accuracy binary     0.76     10  0.0406 pre0_mod02_post0
 3         7 accuracy binary     0.773    10  0.0354 pre0_mod03_post0
 4         9 accuracy binary     0.773    10  0.0220 pre0_mod04_post0
 5        11 accuracy binary     0.740    10  0.0234 pre0_mod05_post0
 6        13 accuracy binary     0.740    10  0.0277 pre0_mod06_post0
 7        15 accuracy binary     0.753    10  0.0234 pre0_mod07_post0
 8        17 accuracy binary     0.76     10  0.0216 pre0_mod08_post0
 9        19 accuracy binary     0.747    10  0.0279 pre0_mod09_post0
10        21 accuracy binary     0.760    10  0.0249 pre0_mod10_post0
```

```
autoplot(knn_cv_single, metric = "accuracy")
```




```
best_k_single <- select_best(knn_cv_single, metric = "accuracy")
best_k_single
```

```
# A tibble: 1 x 2
  neighbors .config
      <dbl> <chr>
1         7 pre0_mod03_post0
```

Model for best k

```
knn_spec_single_best <- nearest_neighbor(neighbors = best_k_single$neighbors, weight_func = '
  set_engine("kknn") |>
  set_mode("classification")

knn_wf_single_best <- workflow() |>
  add_recipe(knn_single_variable_recipe) |>
  add_model(knn_spec_single_best)

single_knn_model <- knn_wf_single_best |>
  fit(data = combined)
```

Model evaluation-accuracy, cv for overfitting

```
single_knn_pred <- predict(single_knn_model, new_data = combined, type = "class") |>
  bind_cols(combined |> select(Democracy)) |>
  mutate(Democracy = factor(Democracy, levels = c("No", "Yes")))

single_knn_cv_best <- collect_metrics(knn_cv_single) |>
  filter(neighbors == best_k_single$neighbors)

knn_single_scores <- tibble(
  model          = "Single variable",
  k              = best_k_single$neighbors,
  cv_accuracy    = single_knn_cv_best$mean,
  cv_std_err     = single_knn_cv_best$std_err,
  train_accuracy = accuracy(single_knn_pred, truth = Democracy, estimate = .pred_class)$.esti
)

knitr::kable(knn_single_scores, digits = 3,
             caption = "Single variable kNN model, cross validation and training")
```

Table 13: Single variable kNN model, cross validation and training

model	k	cv_accuracy	cv_std_err	train_accuracy
Single variable	7	0.773	0.035	0.818

The best k is 7. The model classifies 77.3% of countries correctly in cross validation, against 54.5% for always guessing No. Training accuracy is 81.8%, about 4.5 points higher, so the

model fits the countries it has seen a little better than new ones. This is mild overfitting, and the cross validated number is the fair one.

All variable model

Make a kNN model using all your predictor variables

Model tuning to find k using cv.

```
set.seed(1776)
folds <- vfold_cv(combined, v = 10)

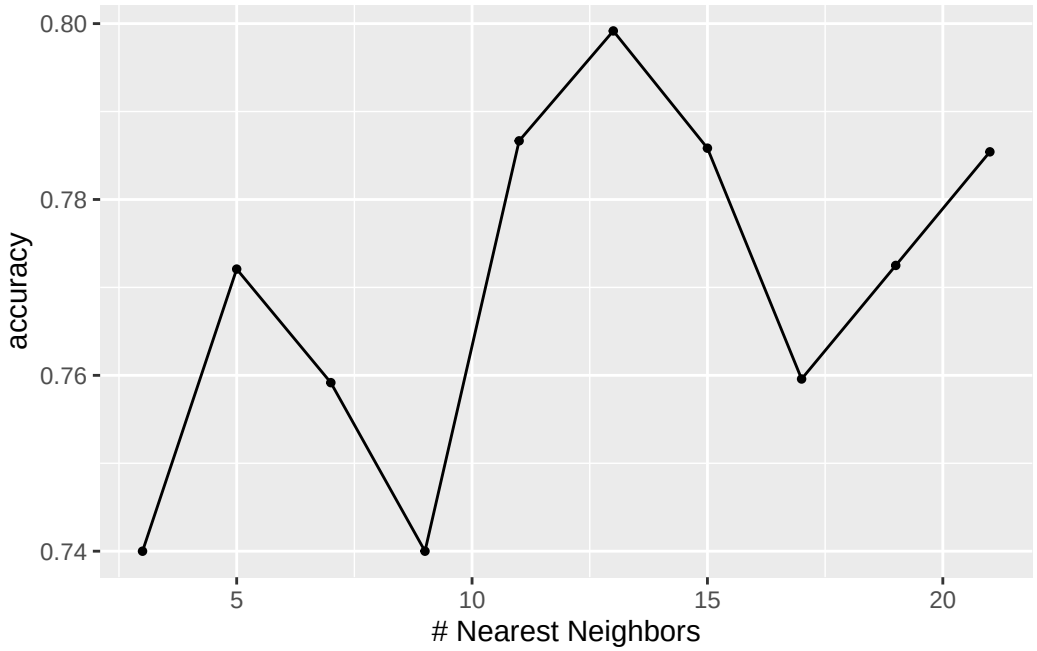
knn_wf_all <- workflow() |>
  add_recipe(knn_all_variable_recipe) |>
  add_model(knn_spec)

knn_cv_all <- tune_grid(
  knn_wf_all,
  resamples = folds,
  grid = knn_grid,
  metrics = class_met
)

collect_metrics(knn_cv_all)
```

```
# A tibble: 10 x 7
   neighbors .metric  .estimator  mean     n std_err .config
       <dbl> <chr>    <chr>      <dbl> <int>   <dbl> <chr>
 1         3 accuracy binary     0.74     10  0.0338 pre0_mod01_post0
 2         5 accuracy binary     0.772    10  0.0355 pre0_mod02_post0
 3         7 accuracy binary     0.759    10  0.0388 pre0_mod03_post0
 4         9 accuracy binary     0.74     10  0.0419 pre0_mod04_post0
 5        11 accuracy binary     0.787    10  0.0323 pre0_mod05_post0
 6        13 accuracy binary     0.799    10  0.0396 pre0_mod06_post0
 7        15 accuracy binary     0.786    10  0.0409 pre0_mod07_post0
 8        17 accuracy binary     0.760    10  0.0352 pre0_mod08_post0
 9        19 accuracy binary     0.772    10  0.0314 pre0_mod09_post0
10        21 accuracy binary     0.785    10  0.0262 pre0_mod10_post0
```

```
autoplot(knn_cv_all, metric = "accuracy")
```




```
best_k_all <- select_best(knn_cv_all, metric = "accuracy")
best_k_all
```

```
# A tibble: 1 x 2
  neighbors .config
      <dbl> <chr>
1        13 pre0_mod06_post0
```

Model for best k

```
knn_spec_all_best <- nearest_neighbor(neighbors = best_k_all$neighbors, weight_func = "recta
  set_engine("kknn") |>
  set_mode("classification")

knn_wf_all_best <- workflow() |>
  add_recipe(knn_all_variable_recipe) |>
  add_model(knn_spec_all_best)

all_knn_model <- knn_wf_all_best |>
  fit(data = combined)
```

Model evaluation-accuracy, cv for overfitting

```
all_knn_pred <- predict(all_knn_model, new_data = combined, type = "class") |>
  bind_cols(combined |> select(Democracy)) |>
  mutate(Democracy = factor(Democracy, levels = c("No", "Yes")))

all_knn_cv_best <- collect_metrics(knn_cv_all) |>
  filter(neighbors == best_k_all$neighbors)

knn_all_scores <- tibble(
  model          = "All variables",
  k              = best_k_all$neighbors,
  cv_accuracy    = all_knn_cv_best$mean,
  cv_std_err     = all_knn_cv_best$std_err,
  train_accuracy = accuracy(all_knn_pred, truth = Democracy, estimate = .pred_class)$.estimat
)

knitr::kable(knn_all_scores, digits = 3,
             caption = "All variable kNN model, cross validation and training")
```

Table 14: All variable kNN model, cross validation and training

model	k	cv_accuracy	cv_std_err	train_accuracy
All variables	13	0.799	0.04	0.838

The best k is 13. The model classifies 79.9% of countries correctly in cross validation. Training accuracy is 83.8%, about 3.9 points higher, which is a similar mild overfit to the single variable model.

Which model was best?

```
bind_rows(knn_single_scores, knn_all_scores) |>
  knitr::kable(digits = 3, caption = "kNN models compared")
```

Table 15: kNN models compared

model	k	cv_accuracy	cv_std_err	train_accuracy
Single variable	7	0.773	0.035	0.818

model	k	cv_accuracy	cv_std_err	train_accuracy
All variables	13	0.799	0.040	0.838

The all variable model has the higher cross validated accuracy, 79.9% against 77.3%. The gap of 2.6 points is smaller than the standard error of about 3.5 to 4 points, so it is not a clear win. It also needs five predictors instead of one and is harder to explain. The extra complexity is not worth it, and the single variable model is the better choice.

Logistic Regression

The response `Democracy` becomes a factor with No first, so each model predicts the probability of Yes. Every model uses the same seed and 10 folds as the kNN models.

```
logit_data <- combined |>
  mutate(
    Democracy = factor(Democracy, levels = c("No", "Yes")),
    income_score = as.numeric(match(
      income_level,
      c("Low income", "Lower middle income", "Upper middle income", "High income")
    )),
    log_people_in_poverty = log(people_in_poverty)
  )

log_spec <- logistic_reg() |>
  set_engine("glm") |>
  set_mode("classification")

set.seed(1776)
logit_folds <- vfold_cv(logit_data, v = 10)

cv_mean <- function(cv, metric) {
  collect_metrics(cv) |>
    filter(.metric == metric) |>
    pull(mean)
}
```

Single variable model

Make a logistic regression model using your best predictor variable. This model uses children per woman, the best predictor from the exploratory analysis.

Model

```
logit_single_wf <- workflow() |>
  add_formula(Democracy ~ children_per_woman) |>
  add_model(log_spec)

logit_single_fit <- logit_single_wf |>
  fit(data = logit_data)

tidy(logit_single_fit) |>
  mutate(odds_ratio = exp(estimate),
         p_value = format.pval(p.value, digits = 2, eps = 1e-4)) |>
  select(term, estimate, std.error, odds_ratio, p_value) |>
  knitr::kable(digits = 3, caption = "Single variable logistic model")
```

Table 16: Single variable logistic model

term	estimate	std.error	odds_ratio	p_value
(Intercept)	3.719	0.646	41.213	<1e-04
children_per_woman	-1.658	0.289	0.191	<1e-04

Each extra child per woman multiplies the odds of being a democracy by $e^{-1.66} \approx 0.19$, a drop of about 81%, and the slope is highly significant with a p-value far below 0.001.

Equation and Graph of model

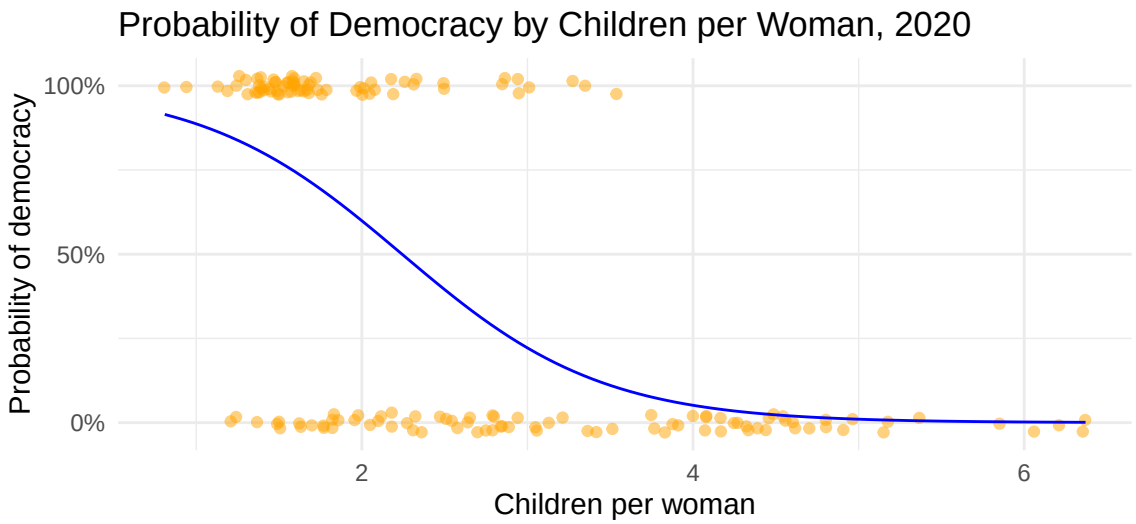
$$\log\left(\frac{p}{1-p}\right) = 3.72 - 1.66 \times \text{children per woman}$$

Here p is the probability that a country is a democracy.

```
curve_data <- tibble(
  children_per_woman = seq(min(logit_data$children_per_woman),
                           max(logit_data$children_per_woman), length.out = 100)
)
curve_data <- bind_cols(curve_data,
                        predict(logit_single_fit, curve_data, type = "prob"))

ggplot() +
  geom_jitter(data = logit_data,
```

```
              aes(children_per_woman, as.numeric(Democracy == "Yes")),
              height = 0.03, alpha = 0.5, color = "orange") +
  geom_line(data = curve_data, aes(children_per_woman, .pred_Yes), color = "blue") +
  scale_y_continuous(breaks = c(0, 0.5, 1), labels = scales::percent) +
  labs(title = "Probability of Democracy by Children per Woman, 2020",
       x = "Children per woman", y = "Probability of democracy",
       caption = "The blue line is the model. Each orange dot is a country, with 0% for No ar
  theme_minimal()
```



The curve falls steadily as families get larger. The model gives a country with 1.5 children per woman about a 77% chance of being a democracy and one with 3 children about a 22% chance. The dots show that countries with few children per woman are mostly democracies and countries with many are mostly not.

Model evaluation-p-values, accuracy, cv for overfitting, roc, auc

```
logit_single_cv <- fit_resamples(
  logit_single_wf,
  resamples = logit_folds,
  metrics = metric_set(accuracy, roc_auc),
  control = control_resamples(save_pred = TRUE)
)

single_pred_class <- predict(logit_single_fit, new_data = logit_data, type = "class")
single_pred_prob  <- predict(logit_single_fit, new_data = logit_data, type = "prob")
```

```
single_train <- logit_data |>
  select(Democracy) |>
  bind_cols(single_pred_class, single_pred_prob)

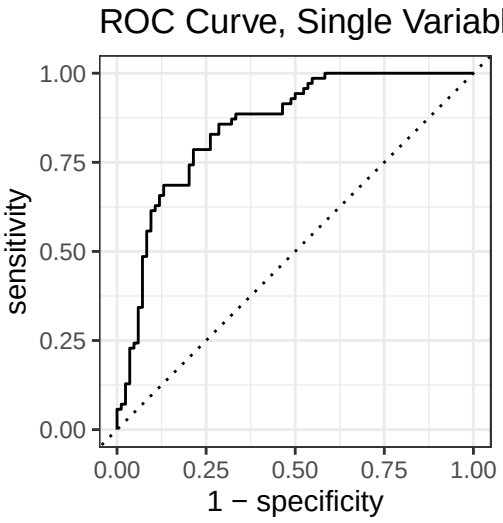
logit_single_scores <- tibble(
  model          = "Single variable",
  cv_accuracy    = cv_mean(logit_single_cv, "accuracy"),
  cv_auc         = cv_mean(logit_single_cv, "roc_auc"),
  train_accuracy = accuracy(single_train, truth = Democracy, estimate = .pred_class)$.estimat
  train_auc      = roc_auc(single_train, truth = Democracy, .pred_Yes,
                           event_level = "second")$.estimate
)

knitr::kable(logit_single_scores, digits = 3,
             caption = "Single variable logistic model, cross validation and training")
```

Table 17: Single variable logistic model, cross validation and training

model	cv_accuracy	cv_auc	train_accuracy	train_auc
Single variable	0.772	0.857	0.773	0.855

```
collect_predictions(logit_single_cv) |>
  roc_curve(truth = Democracy, .pred_Yes, event_level = "second") |>
  autoplot() +
  labs(title = "ROC Curve, Single Variable Logistic Model",
       caption = "Predictions from 10 fold cross validation.")
```



Predictions from 10 fold cross validation.

The p-value for children per woman is far below 0.001, so it is a clearly useful predictor. The model classifies 77.2% of countries correctly in cross validation, against 54.5% for always guessing No, and the cross validated AUC is 0.857. Training accuracy of 77.3% and AUC of 0.855 are almost the same, so there is no sign of overfitting.

All variable model

Make a logistic regression model using all your predictor variables

Model

Income group is scored 1 to 4 from Low to High income in this model, and people in poverty is logged because its counts are extremely skewed. Income group is scored as a number because separate categories break the model. No low income country is a democracy, so the Low income estimate runs off to about -21 with a standard error above 1,300.

```
suppressWarnings(glm(Democracy ~ income_level, data = logit_data, family = binomial)) |>
  tidy() |>
  knitr::kable(digits = 3, caption = "Income group entered as separate categories")
```

Table 18: Income group entered as separate categories

term	estimate	std.error	statistic	p.value
(Intercept)	2.225	0.526	4.227	0.000
income_levelLow income	-20.791	1331.428	-0.016	0.988

term	estimate	std.error	statistic	p.value
income_levelLower middle income	-3.438	0.628	-5.470	0.000
income_levelUpper middle income	-2.078	0.612	-3.393	0.001

```
logit_all_wf <- workflow() |>
  add_formula(Democracy ~ income_score + sanitation_access_pct + children_per_woman +
                co2_per_capita + log_people_in_poverty) |>
  add_model(log_spec)

logit_all_fit <- logit_all_wf |>
  fit(data = logit_data)

tidy(logit_all_fit) |>
  mutate(odds_ratio = exp(estimate),
         p_value = format.pval(p.value, digits = 2, eps = 1e-4)) |>
  select(term, estimate, std.error, odds_ratio, p_value) |>
  knitr::kable(digits = 3, caption = "All variable logistic model")
```

Table 19: All variable logistic model

term	estimate	std.error	odds_ratio	p_value
(Intercept)	0.065	3.134	1.067	0.98342
income_score	2.464	0.556	11.749	< 1e-04
sanitation_access_pct	-0.052	0.020	0.950	0.00848
children_per_woman	-1.871	0.485	0.154	0.00012
co2_per_capita	-0.111	0.072	0.895	0.12332
log_people_in_poverty	0.162	0.113	1.176	0.15257

```
predictor_names <- c("income_score", "sanitation_access_pct", "children_per_woman",
                     "co2_per_capita", "log_people_in_poverty")

tibble(
  predictor = predictor_names,
  vif = map_dbl(predictor_names, \(v) {
    r2 <- summary(lm(reformulate(setdiff(predictor_names, v), v),
                     data = logit_data))$r.squared
    1 / (1 - r2)
  })
) |>
  knitr::kable(digits = 2, caption = "Variance inflation factors")
```

Table 20: Variance inflation factors

predictor	vif
income_score	3.98
sanitation_access_pct	4.02
children_per_woman	4.11
co2_per_capita	2.34
log_people_in_poverty	2.44

Holding the other predictors fixed, each step up in income group multiplies the odds of democracy by about 12 and each extra child per woman multiplies them by about 0.15. Each extra percentage point of sanitation access multiplies them by about 0.95. CO2 per capita and logged poverty are not significant, with p-values of 0.12 and 0.15, so they add little. Sanitation has a negative sign even though democracies have higher sanitation on their own. It overlaps with income group and children per woman, with variance inflation factors near 4, so its slope should not be read alone.

Equation of model

$$\log\left(\frac{p}{1-p}\right) = 0.07 + 2.46 \times \text{income score} - 0.052 \times \text{sanitation} - 1.87 \times \text{children per woman} - 0.111 \times \text{CO2} + 0.162 \times \log(\text{poverty})$$

Model evaluation-p-values, accuracy, cv for overfitting, roc, auc

```
logit_all_cv <- fit_resamples(
  logit_all_wf,
  resamples = logit_folds,
  metrics = metric_set(accuracy, roc_auc),
  control = control_resamples(save_pred = TRUE)
)

all_pred_class <- predict(logit_all_fit, new_data = logit_data, type = "class")
all_pred_prob  <- predict(logit_all_fit, new_data = logit_data, type = "prob")

all_train <- logit_data |>
  select(Democracy) |>
  bind_cols(all_pred_class, all_pred_prob)

logit_all_scores <- tibble(
```

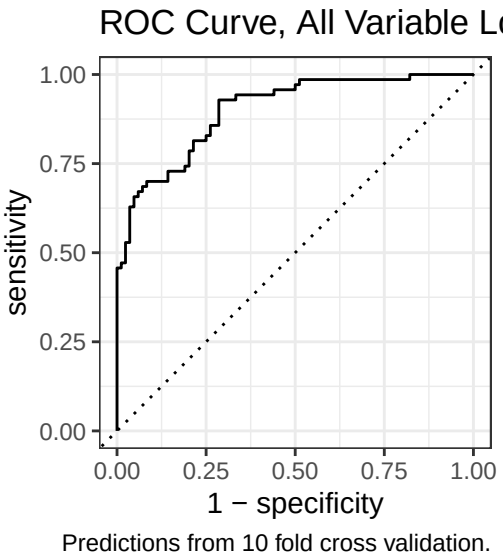
```
  model          = "All variables",
  cv_accuracy    = cv_mean(logit_all_cv, "accuracy"),
  cv_auc         = cv_mean(logit_all_cv, "roc_auc"),
  train_accuracy = accuracy(all_train, truth = Democracy, estimate = .pred_class)$.estimate,
  train_auc      = roc_auc(all_train, truth = Democracy, .pred_Yes,
                           event_level = "second")$.estimate
)

knitr::kable(logit_all_scores, digits = 3,
             caption = "All variable logistic model, cross validation and training")
```

Table 21: All variable logistic model, cross validation and training

model	cv_accuracy	cv_auc	train_accuracy	train_auc
All variables	0.787	0.899	0.825	0.918

```
collect_predictions(logit_all_cv) |>
  roc_curve(truth = Democracy, .pred_Yes, event_level = "second") |>
  autoplot() +
  labs(title = "ROC Curve, All Variable Logistic Model",
       caption = "Predictions from 10 fold cross validation.")
```



Income group, children per woman and sanitation are significant, so they belong in the model, while CO2 and poverty could be dropped. The model classifies 78.7% of countries correctly in cross validation with an AUC of 0.899. Training accuracy is 82.5% and training AUC is 0.918,

so the model fits the training data a little better than new data. This is mild overfitting, and the cross validated numbers are the fair ones.

Which model was best?

```
bind_rows(logit_single_scores, logit_all_scores) |>
  knitr::kable(digits = 3, caption = "Logistic models compared")
```

Table 22: Logistic models compared

model	cv_accuracy	cv_auc	train_accuracy	train_auc
Single variable	0.772	0.857	0.773	0.855
All variables	0.787	0.899	0.825	0.918

The all variable model ranks countries better, with a cross validated AUC of 0.899 against 0.857. Its accuracy is only 1.5 points higher at 78.7% against 77.2%, which is within the standard error of about 3 points. It also overfits a little more, with a gap of 3.8 points between training and cross validated accuracy against about 0.1 points for the single variable model, and two of its five predictors are not significant. The extra complexity is only partly worth it. It helps rank countries by how likely they are to be democracies, but it adds little for a simple yes or no call, so the single variable model is the simpler choice and a model with only income, children per woman and sanitation would be worth trying.

Best model

Which model would you reccomend the lab use? Why? Are there tradeoff between interpretability and model metrics?

Your question

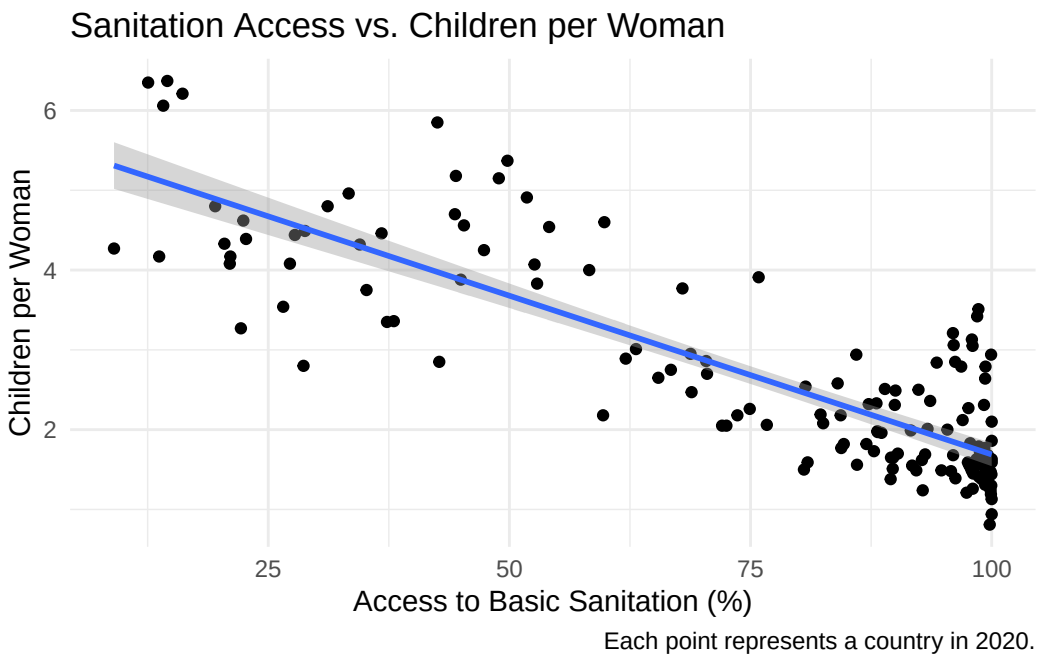
Is access to basic sanitation associated with the average number of children per woman across countries?

Discussion

Access to basic sanitation is an important indicator of living conditions and public health. We wanted to investigate whether countries with greater access to basic sanitation tend to have lower fertility rates. Comparing these two variables may help us understand how different aspects of development are associated across countries.

```
combined |>
  ggplot(aes(x = sanitation_access_pct, y = children_per_woman)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Sanitation Access vs. Children per Woman",
    x = "Access to Basic Sanitation (%)",
    y = "Children per Woman",
    caption = "Each point represents a country in 2020."
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```r
combined |>
  summarise(
    correlation = cor(sanitation_access_pct, children_per_woman)
  )
```

```
# A tibble: 1 x 1
  correlation
        <dbl>
1      -0.846
```

Answer

The visualization shows a strong negative relationship between access to basic sanitation and the number of children per woman. Countries with higher access to basic sanitation generally have fewer children per woman, while countries with lower sanitation access tend to have higher fertility rates. The correlation coefficient of -0.846 supports this observation, indicating a strong negative association between the two variables. However, this relationship does not establish causation, and other factors, such as income, education, and healthcare access, may also contribute to the observed pattern.